NETFLIX AFTER HOURS

OPERATIONAL TELEMETRY & SLO DASHBOARD SPECIFICATION

Document Version: 1.0-FINAL
Real-Time SRE Architecture

1. TELEMETRY INGESTION & MONITORING ARCHITECTURE

The **After Hours Operational Dashboard** processes high-frequency edge events (playback state shifts, audio DSP curves, biometric events) via a streaming telemetry pipeline to monitor network egress savings, active session state transitions, and system error budgets in real time.

FIGURE 1.1: REAL-TIME TELEMETRY & ALERTING PIPELINE



2. CORE TELEMETRY METRICS SCHEMA

Metric Key	Type	Description & Dimensions	Target Value / Range
`egress_bandwidth_gbps`	Gauge	CDN egress bandwidth by region, hour, and experiment cell.	**-12% to -18%** (11 PM - 3 AM)
`active_sleep_fades_count`	Gauge	Concurrent active display dimming or audio decay sequences.	Scales with late-night traffic
`sleep_cancel_events_total`	Counter	User taps/keypresses canceling sleep fade sequence.	**< 2.0%** of active fades
`decay_bookmark_sync_latency_ms`	Histogram	Time to sync local bookmark to cloud EVCache/Cassandra.	**p99 < 150 ms**
`wearable_telemetry_errors_total`	Counter	Failed Apple/Health Connect/Oura token validations.	**< 0.05%** error rate

3. KEY PROMQL / ATLAS TELEMETRY QUERIES

1. Late-Night CDN Bandwidth Savings Delta (%):

```
( sum(rate(egress_bandwidth_gbps{cell="control"}[5m])) - sum(rate(egress_bandwidth_gbps{cell="variant_c"}[5m])) ) / sum(rate(egress_bandwidth_gbps{cell="control"}[5m])) * 100
```

2. False-Positive Cancellation Rate (Circuit Breaker Input):

```
sum(rate(sleep_cancel_events_total[5m])) / sum(rate(active_sleep_fades_initiated_total[5m])) * 100
```

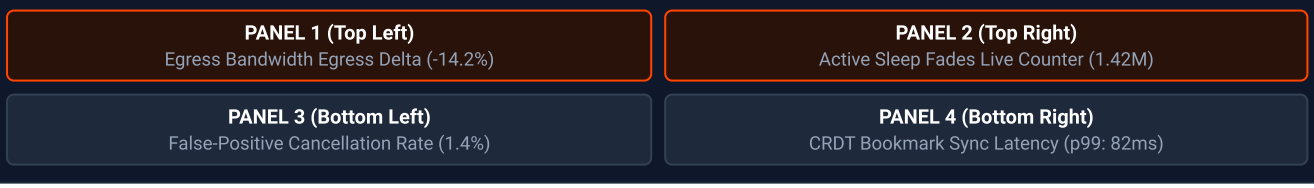
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GRAFANA PANEL WIREFRAME, SLOS & ALERT ROUTING

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SRE Alerting Matrix

4. DASHBOARD VISUAL PANEL LAYOUT STRUCTURE

FIGURE 4.1: GRAFANA DASHBOARD GRID LAYOUT WIREFRAME



5. SERVICE LEVEL OBJECTIVES (SLOS) & ERROR BUDGET ALLOCATIONS

Service Metric	SLI Target	30-Day Error Budget	Burn Rate Action
Decay Sync Availability	99.9% Successful Syncs	0.1% Failed Syncs (43.2 min)	Page SRE if 14.4x burn in 1h
Audio DSP Fade Accuracy	99.95% Linear/Log Compliance	0.05% Distortion Events	Halt audio rollouts if burn > 5%
Wearable Latency Window	p95 < 30 seconds	5% Delayed Telemetry Tokens	Fallback to manual timers
Circuit Breaker Trigger	100% Automated Trip < 30s	0 Missed Breach Events	P0 Immediate Escalation

6. PAGERDUTY & KAFKA ALERT THRESHOLD RULES

Severity	Condition / Threshold	Automated System Action	Notification Channel
P0 - CRITICAL	False-positive cancels > 3.5% OR error rate > 0.1% over 5m.	**Trip LaunchDarkly Kill-Switch** to Control in < 30s.	PagerDuty On-Call + Slack `#afterhours-p0`
P1 - HIGH	Bookmark sync p99 latency > 300 ms for 10 minutes.	Enable local SQLite fallback write queue.	PagerDuty SRE Primary
P2 - MEDIUM	Wearable API authentication failure rate > 1.0%.	Fallback Wearable cell traffic to scheduled timers.	Slack `#afterhours-alerts`